BISCUIT: Causal Representation Learning from Binary Interactions

Nicola Annunziata · Marco Barberio · Nandani Jagdishbhai Kathiriya · Niccolò Maestripieri · Michele Panicucci

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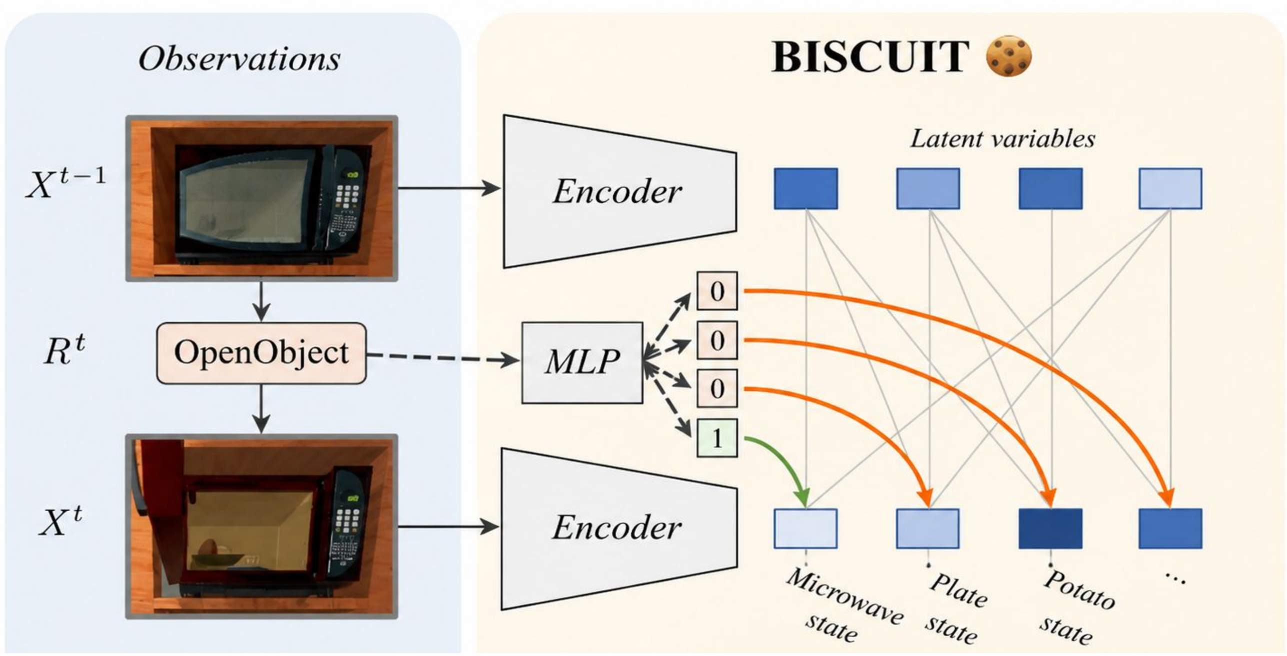
UNIVERSITÀ DI PISA

1. Introduction

Problem: Learning a low-dimensional representation of an environment is crucial in robotics, embodied AI, and reinforcement learning. A promising direction is causal representation learning, which aims to identify the underlying causal variables and their relations from high-dimensional observations, e.g., images. However, agents interact with the world but their interaction targets remain unknown.

Gap: Prior methods either require known intervention targets (CITRIS), need at least 2K+1 regime values (iVAE, LEAP), or fail on additive Gaussian noise models (DMS restricts the causal graph structure).

Contribution: BISCUIT proves that causal variables are identifiable (up to permutation and element-wise transforms) when each variable has two different mechanisms — observational and interventional — described by a distinct binary interaction pattern. It jointly learns causal variables and their unknown binary interaction targets — with as few as $\lfloor \log_2(K) \rfloor + 2$ distinct actions, without supervision.

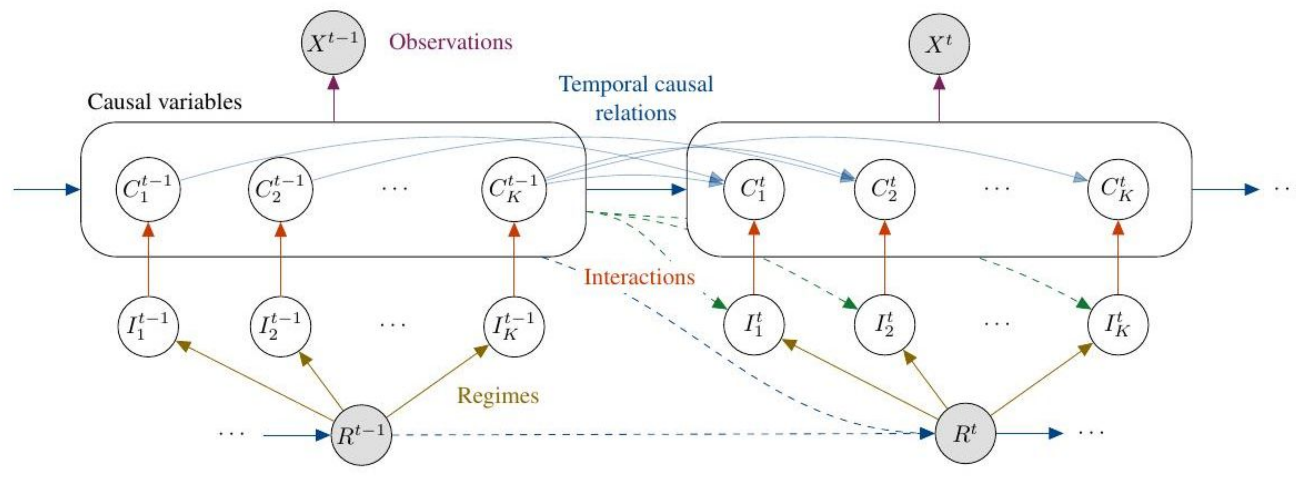


BISCUIT learns causal variables by encoding R^t as binary interactions $I_i \in \{0,1\}$.

2. Approach

Estimated Causal Model: An estimated model identifies the true causal model if:

- **(Observations)** The estimated model induces the same likelihood as the true model.
- **(Distinct Interaction Patterns)** Each causal variable in the true model has a distinct interaction pattern.
- One of the following holds:
  - **(Dynamic Variability)** Each variable's log-likelihood difference is twice differentiable and its second derivative is not identically zero.
  - **(Time Variability)** For any set of causal variables at a given timestamp, there exist K+1 different values of the causal variables at the previous timestamp such that the vectors containing the gradients of the log-likelihood differences with respect to each causal variable are linearly independent.



Causal model: C_i^t evolve via binary I_i^t driven by R^t; X^t is their entangled observation.

Architecture: BISCUIT is a VAE (encoder/decoder + structured prior) which learns the causal variables and the agent's binary interactions with them in an unsupervised manner. The prior uses per-variable MLPs: one predicts I_i from (R_t, z_t^{-1}) via temperature-scaled tanh; another predicts $p(z_{it} \mid z_t^{-1}, I_i)$. BISCUIT-NF separates autoencoder and normalizing-flow stages for visually complex scenes.

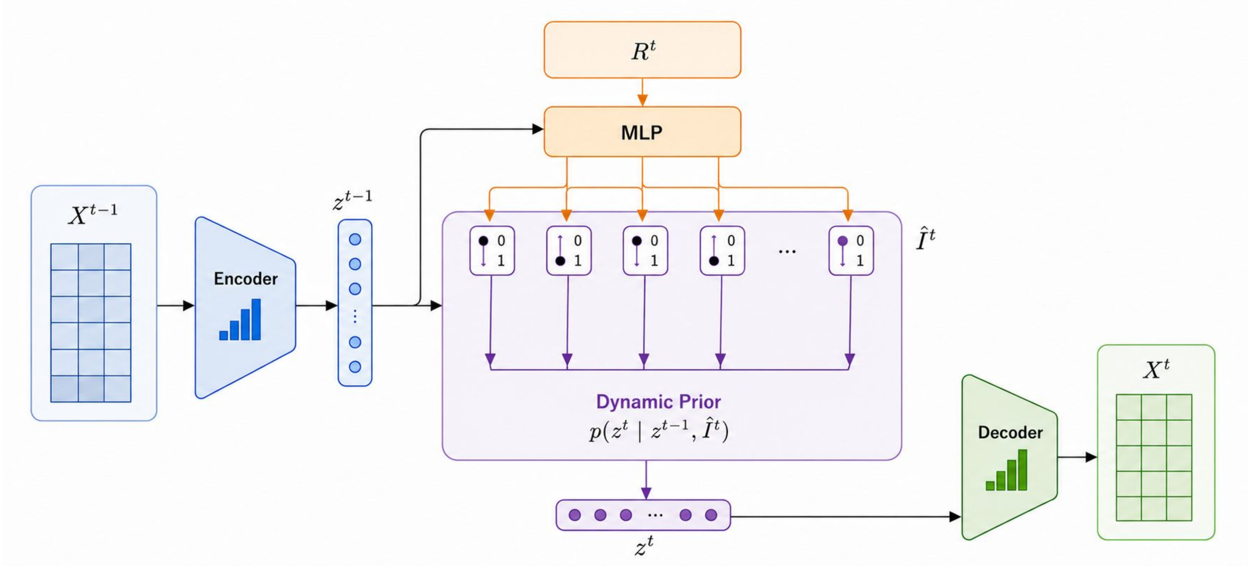
Biscuit VAE: A Variational autoencoder-based version of BISCUIT where the encoder maps observations into a lower-dimensional latent space and the decoder reconstructs the input images.

Biscuit NF: A BISCUIT variant that combines the autoencoder with a normalizing flow. The normalizing flow transforms the latent representations learned by the autoencoder into disentangled causal variables while modeling the temporal dynamics and interactions between variables.

Loss:

$$\mathcal{L}^t = -\mathbb{E}_{q_\phi(z^t|x^t)}\left[\log p_\theta(x^t|z^t)\right] + \mathbb{E}_{q_\phi(z^{t-1}|x^{t-1})}\left[\mathrm{KL}\left(q_\phi(z^t|x^t)\|p_\omega(z^t|z^{t-1}, R^t)\right)\right]$$

Model Prior The prior models the temporal evolution of each latent variable independently. For every latent dimension, an MLP predicts a binary interaction variable from the regime variable and previous latent state. Another MLP uses this interaction variable together with the previous latent representation to predict the distribution of the next latent state.



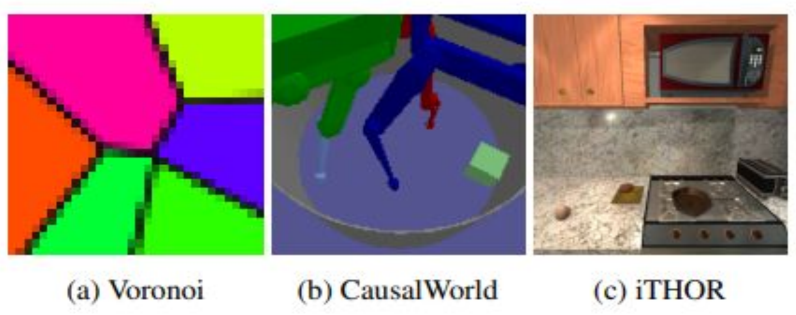
BISCUIT architetture

3. Experiments

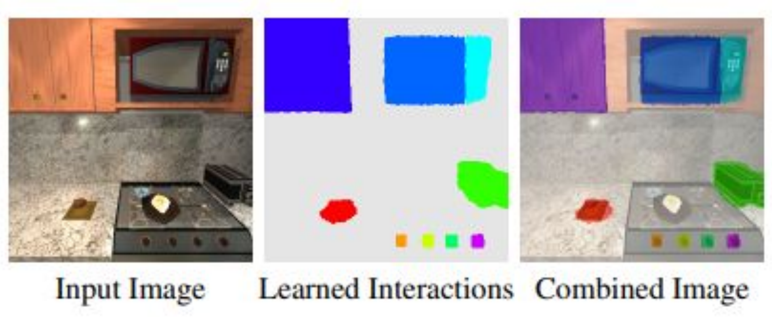
Datasets:
1. **Voronoi:** Synthetic benchmark, additive Gaussian noise, 6 and 9 causal variables, robotic arm as regime;
2. **CausalWorld:** Tri-finger robot manipulating a cube in 3D. Tests robustness, as real interactions are not strictly binary (depend on velocity, rotation, etc.).
3. **iTHOR:** Realistic environment with 18 causal variables (embodied AI). Conditional variable states depends on previous variables interaction (time dependency)

Metric & baselines:
R^2-diag (higher=better) measures alignment of learned latents to true causal variables; R^2-sep (lower=better) measures spurious correlations. Compared against iVAE, LEAP, and DMS.

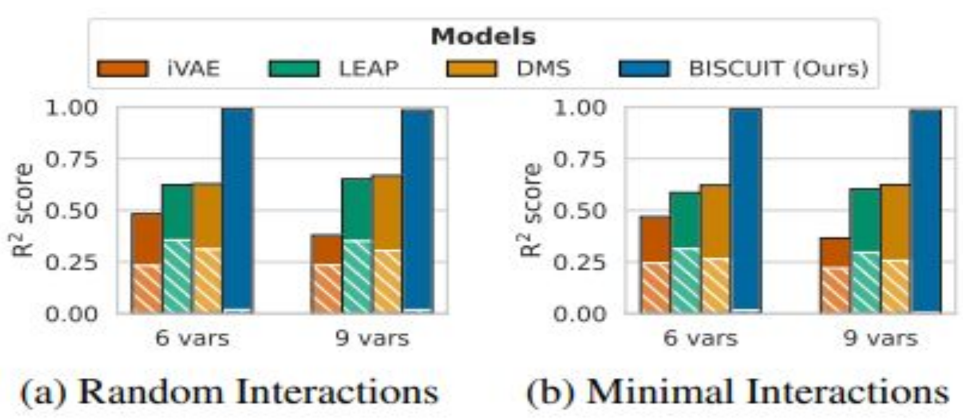
Results:
1. **Voronoi:** BISCUIT reliably recovers causal variables even with the minimum interaction regimes.
2. **CausalWorld:** BISCUIT-NF does not only reconstruct the image, but also separates the main causal variables. It also learns to recognize whether a robot-cube collision occurs.
3. **iTHOR:** BISCUIT mainly struggles with egg and plate because they are highly correlated in the environment. However, it shows a modular representation, because it can combine latents from two images and generate new states, such as raw egg + stove on, generating a novel state.

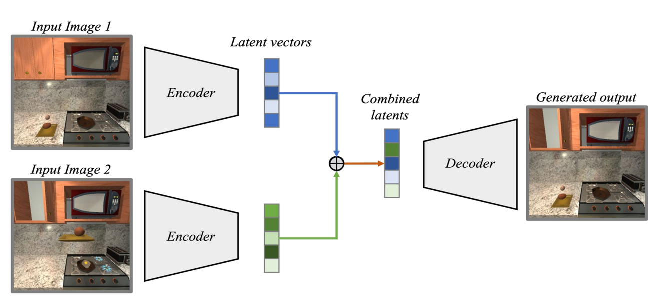


Three environments with increasing complexity



Learned Interaction Variables





Example of novel state being generated by BISCUIT

Models	CausalWorld	iTHOR
iVAE (Khemakhem et al., 2020a)	0.28 / 0.00	0.48 / 0.35
LEAP (Yao et al., 2022b)	0.30 / 0.00	0.63 / 0.45
DMS (Lachapelle et al., 2022b)	0.32 / 0.00	0.61 / 0.40
BISCUIT-NF (Ours)	**0.97** / 0.01	**0.96** / **0.15**

R^2 scores for the identification of causal variables

Takeaway:
BISCUIT demonstrates that binary interactions make causal variables identifiable from high-dimensional observations without known intervention targets.

4. Personal considerations

Reflection: BISCUIT presents an elegant approach for causal representation learning in interactive environments. Unknown binary interactions can make latent causal variables identifiable, even from high-dimensional observations such as images. It's particularly interesting how the model combines temporal dynamics, causal assumptions, and variational latent representations within a unified framework.

Assumptions and Limitations: The method assumes that interactions can be represented through binary variables, but many real-world interactions are continuous and more complex. Second, the identifiability guarantees require distinct interaction patterns between causal variables. In practice, highly correlated variables or simultaneous interactions may violate this assumption. Another limitation is scalability: the model performs well on moderate-sized environments, but larger real-world scenes with many interacting objects could significantly increase training complexity. Finally, the theoretical guarantees depend on assumptions such as causal sufficiency and specific temporal structures, which may not always hold in real applications.

Responsible AI: Causal representations learned without supervision could encode spurious correlates from biased training environments. Deploying such models in robotics or embodied AI requires careful auditing of what variables are learned and whether they reflect true causal structure.

Lessons learned: Binary structure is a surprisingly powerful inductive bias, it breaks the rotational invariance that makes additive Gaussian noise models hard.

Future directions:

- Continuous interaction variables
- Partial identifiability settings
- Larger embodied AI environments
- Real-world robotic applications

Lippe P., Magliacane S., Löwe S., Asano Y.M., Cohen T., Gavves E. — BISCUIT: Causal Representation Learning from Binary Interactions. UAI 2023, PMLR 216:1263–1273.